

Abhishek Raj

B.Tech, Computer Science and Engineering | Indian Institute of Technology, Indore

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GitHub [↗](#) | LinkedIn [↗](#) | Portfolio [↗](#)

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology Indore	7.52 (Current)	2024-Present
Senior Secondary	CBSE Board	92.2%	2024
Secondary	CBSE Board	93.4%	2022

PROJECTS

- **Trip Buddy – Complete Trip Planner Application, Play Store** [↗](#) Jun. 2025 – Jul. 2025
Kotlin Multiplatform, Jetpack Compose, Spring Boot, MySQL, Firebase [GitHub](#) [↗](#)
 - Engineered a cross-platform travel app and **deployed it live on Google Play Store** using **Kotlin Multiplatform (KMP)** with **Jetpack Compose** targeting – **Android and iOS** across **20+ screens**; built **Spring Boot** backend with **MVC architecture** (Controller /Service/Repository) and **Ktor** for API communication with **<155ms** response latency.
 - Aggregated real-time data from **8 external APIs** (Amadeus, OpenWeatherMap, OpenTripMap) via async **Ktor** client, delivering flights, weather, and attractions in a single session with **<3.5s** latency.
 - Secured auth via **Firebase Google Sign-In**; containerized **Spring Boot** services with **Docker** and automated deployments via **GitHub Actions CI/CD** to **Aiven cloud MySQL**, enabling **zero-downtime** releases.
- **CoderBuddy – Multi-Agent AI Code Generator, Website** [↗](#) Mar. 2026 – Apr. 2026
FastAPI, LangGraph, React, TypeScript, Monaco Editor [GitHub](#) [↗](#)
 - Architected a **4-agent LangGraph** pipeline (Planner → Architect → Coder → Reviewer) across **6 Pydantic schemas** with **dynamic routing** and **self-correction cycles**, achieving **90% pass rate** on a **10-prompt benchmark**.
 - Implemented an **LLM output runner** with **two-stage fallback** (schema enforcement → regex JSON parsing) across **3 agents**, raising tool-call success rate from **93.3% to 98.2%** and eliminating pipeline crashes on schema violations.
 - Engineered a **zero-dependency live preview** in **React**, inlining **CSS/JS** into sandboxed **HTML Blobs** rendered in an **iframe** with worst-case render latency of **3.2ms**, eliminating external CDN calls at runtime.
- **Shop4Ever – Supermarket Management System, Website** [↗](#) Aug. 2025 – Nov. 2025
React, Node.js, Express.js, MySQL, Brevo API, Tailwind CSS, ImageKit [GitHub](#) [↗](#)
 - Built a **full-stack retail platform** with **LLM-powered shopping recommendations** across **3 role-based dashboards** (admin/staff/customer), secured via **JWT + Bcrypt + OTP** authentication.
 - Engineered a **12-entity 3NF MySQL** schema with **ACID transactions**, supporting **50 concurrent orders** with **zero race conditions**; implemented **SQL triggers** for automated inventory synchronization in **<150ms**.
 - Optimized MySQL queries using **B+ Tree indexes** and **selection pushdown**, reducing execution time by **~40%**.

TECHNICAL SKILLS

- **Languages:** C++, Kotlin, JavaScript, TypeScript, SQL, Python
- **Frameworks/Tools:** Jetpack Compose, KMP, Firebase, React, HTML/CSS, Spring Boot, Node.js, Express.js, RESTful APIs, Git, GitHub Actions (CI/CD), Postman, Docker, LangGraph, LangChain
- **Databases:** MySQL, MongoDB*, PostgreSQL * Elementary proficiency

KEY COURSES TAKEN

- **Mathematics:** Linear Algebra, Complex Analysis, Calculus, Discrete Maths, Probability & Random Processes, Mathematics in AIML, Numerical Methods
- **CS Core:** Data Structure & Algorithm, Design & Analysis of Algorithms, OOP in C++, DBMS, Logic Design, Automata Theory & Logic, Computer Architecture, Foundation of Hardware Security

POSITIONS OF RESPONSIBILITY

- **App Development Head**, Google Developer Group, IIT Indore Nov. 2025 – Present
- **Technicals Head**, Gaming Club, IIT Indore Oct. 2025 – Present
- **Software Division Member**, Programming Club, IIT Indore Feb. 2026 – Present
- **Coordinator**, CSESA, IIT Indore Aug. 2025 – Present

ACHIEVEMENTS

- **Winner**, Ten3 Hackathon – Google Developer Group, IIT Indore Nov. 2024
- **LeetCode Rating 1556**, on LeetCode [↗](#) Present
- **Codeforces Specialist**, Max Rating 1485 on Codeforces [↗](#) Present
- **Finalist – Technocrats Hackathon by Quanta Crew**, Selection Mail [↗](#) , Certificate [↗](#) Dec. 2025
- **Finalist**, IIT Indore Summer of Code – Software Division Aug. 2025
- **Bronze — Fashion Show**, Inter IIT Culturals Meet Dec. 2025
- **Winner**, Tech-Eshi's Castle – Robotics Club IIT Indore Nov. 2024